

CAPE TOWN PORTNET, South Africa

WMO: 688170

Lat: 33.9206S

Lon: 18.4431E

Elev: 7

StdP: 14.69

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	48.6	50.3	32.6	27.0	65.7	35.1	29.9	62.4	31.5	58.0	28.2	57.2	3.6	240	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.0	89.7	66.8	85.8	65.9	82.6	65.0	70.1	84.6	68.8	82.7	67.6	80.8	10.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.2	93.1	76.8	63.8	88.5	74.7	62.6	85.0	73.5	34.1	85.0	32.9	82.5	32.0	80.9	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.3	30.5	27.1	45.6	97.0	1.7	4.4	44.4	100.2	43.4	102.8	42.5	105.3	41.3	108.5		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.0	71.8	72.0	69.4	66.3	62.1	59.0	58.7	58.9	60.9	64.7	66.5	69.9
	DBStd	6.66	4.91	4.94	5.26	5.41	4.02	3.36	3.84	3.66	4.53	5.03	4.88	4.73
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1000	3	3	13	45	110	183	199	195	141	67	34	7
	CDD50	5473	677	617	601	488	376	271	271	276	327	455	496	618
	CDD65	998	215	200	149	83	21	4	5	5	18	57	81	160
	CDH74	4856	1131	1036	706	399	80	20	24	24	84	278	334	740
	CDH80	1362	336	302	209	126	11	2	2	2	20	73	82	197
(14) Wind	WSAvg	9.8	13.1	11.9	10.4	8.8	6.5	6.9	7.4	7.7	9.7	10.9	12.5	12.3
(15) Precipitation	PrecAvg	22.6	0.7	0.6	0.9	1.7	3.0	4.3	3.2	3.4	1.8	1.3	0.9	0.7
	PrecMax	32.6	3.0	2.6	3.4	6.8	7.8	9.8	6.2	9.1	4.4	5.3	3.1	2.1
	PrecMin	13.7	0.0	0.0	0.0	0.2	0.6	0.9	0.0	0.8	0.2	0.0	0.0	0.0
	PrecStd	4.7	0.7	0.7	0.9	1.2	1.9	2.2	1.6	2.0	1.1	1.1	0.8	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.0	93.5	92.7	90.4	81.3	77.1	77.2	76.9	82.1	88.5	87.6	91.2
		MCWB	69.4	68.1	67.4	65.2	60.1	57.7	57.5	59.0	59.9	64.5	62.8	69.8
	2%	DB	87.7	87.3	85.3	82.9	75.7	71.4	71.6	71.4	75.2	80.9	81.6	84.9
		MCWB	68.1	67.3	66.2	63.4	59.4	55.1	56.7	57.2	57.9	61.9	62.4	66.4
	5%	DB	83.4	83.2	80.9	77.8	71.7	67.6	67.8	67.4	70.6	76.4	77.4	81.1
		MCWB	66.7	66.1	65.3	62.0	59.1	55.3	56.1	56.0	56.9	61.0	61.3	65.2
	10%	DB	79.9	79.8	77.4	74.0	68.6	64.3	64.7	64.4	67.1	72.5	74.1	77.8
		MCWB	66.3	65.7	64.5	61.1	58.1	55.3	55.1	55.0	56.1	59.9	60.4	64.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.1	72.2	70.1	67.7	63.5	60.6	60.5	61.2	61.3	67.3	66.6	71.2
		MCDB	87.8	85.6	84.6	84.6	74.4	69.1	72.8	71.5	79.4	87.0	83.6	90.0
	2%	WB	70.3	69.5	68.4	65.7	61.7	58.6	58.9	59.3	59.5	64.8	64.4	68.8
		MCDB	84.4	82.3	81.2	81.8	70.8	64.8	67.3	69.8	71.7	79.6	77.5	84.0
	5%	WB	68.9	67.9	66.7	64.1	60.5	57.6	57.7	57.9	58.4	62.8	63.1	67.3
		MCDB	82.4	80.9	78.8	77.4	68.7	62.9	65.1	66.4	67.8	75.3	74.9	80.7
	10%	WB	67.5	66.7	65.3	62.7	59.3	56.7	56.7	56.4	57.3	61.3	62.1	66.1
		MCDB	80.1	79.0	76.1	73.5	66.4	61.6	63.1	63.1	64.9	72.1	73.1	77.9
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	13.0	12.6	11.8	10.6	10.0	10.4	10.1	9.8	11.2	11.0	11.9
		MCDBR	19.6	19.6	19.4	20.1	17.7	17.6	17.6	16.9	18.0	19.4	17.8	18.9
		MCWBR	8.0	7.7	7.6	7.4	8.0	9.0	9.0	8.4	7.8	7.9	7.7	8.4
	5% WB	MCDBR	19.2	17.5	18.0	19.8	14.8	12.3	14.1	15.8	16.1	18.8	17.1	19.3
		MCWBR	8.2	7.6	7.5	7.9	7.6	7.1	7.7	8.3	7.3	7.7	7.8	8.4
(40) Clear-Sky Solar Irradiance	taub		0.343	0.345	0.336	0.323	0.322	0.317	0.319	0.334	0.344	0.353	0.340	0.344
	taud		2.540	2.543	2.565	2.593	2.589	2.576	2.566	2.521	2.456	2.446	2.526	2.528
	Ebn at Noon		315	308	300	286	269	261	267	278	292	302	314	316
	Edh at Noon		35	33	30	26	23	22	24	28	33	36	35	36
(44) All-Sky Solar Radiation	RadAvg		2606	2326	1866	1363	942	792	884	1121	1549	2045	2430	2634
	RadStd		58	68	71	68	60	46	53	54	62	95	71	61

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	+0.65	+0.88	N/A	N/A	+0.95	+0.85	-4	-133	+200	+110	

Nomenclature: See separate page